Amazon Sales Analysis

Project Report | Amazon.in Order Data

Project Overview

This project examines Amazon.in sales/order data to understand sales performance, order statuses, product-category demand, fulfilment methods, shipping preferences, and geographic patterns. The objective is to convert raw order-level records into clear business insights and dashboard-ready metrics.

Project Objectives

• Measure overall order activity, quantity, and available Amount values.
• Review cancellations, returns, and other order statuses.
• Identify high-volume product categories and shipping preferences.
• Compare Amazon and Merchant fulfilment patterns.
• Explore date and geographic trends and present results visually.

Dataset Summary

Attribute	Details
File	Amazon Sale Report(2).csv
Records	128,975 rows
Fields	24 columns
Currency	INR where available
Period	April 2022 onward, based on the Date field

Tools and Workflow

Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, and Power BI. The workflow covers loading the CSV, checking data quality, reviewing missing values and duplicates, cleaning types/fields, conducting categorical and numerical EDA, visualizing patterns, and summarizing findings in a dashboard.

Key Metrics from the Uploaded File

Metric	Value
Total records	128,975
Columns	24
Sum of non-missing Amount values	■78,592,678.30

Rows with Amount present	121,180
Total Qty	116,649
Cancelled records	18,332 (about 14.21%)
Amazon fulfilment share	About 69.55%
Expedited shipping share	About 68.71%

Important interpretation: Amount is summed across all rows where Amount is present. Because the dataset includes cancelled and other non-completed statuses, this total should not automatically be treated as net realized revenue. Define eligible statuses before reporting revenue.

Initial Findings

• Set and kurta are the two largest categories by record count.
• Amazon fulfilment accounts for about 69.55% of records; Merchant fulfilment accounts for about 30.45%.
• Expedited shipping accounts for about 68.71% of records.
• There are 7,795 missing Amount values; investigate them rather than filling them blindly with zero, mean, or median.
• The file contains an Unnamed: 22 field that appears extra/empty and should be checked.
• A full-row duplicate check found no exact duplicate rows. Repeated Order IDs may still be valid at item/record level.

Recommended Visuals

• KPI cards: total records, available Amount, total quantity, average order value, cancellation rate.
• Category-wise record count and Amount.
• Order-status distribution and cancellation/return analysis.
• Amazon vs Merchant fulfilment and Standard vs Expedited shipping.
• Monthly/date trends and state/city breakdowns.
• Quantity-versus-Amount analysis.

Business Recommendations

• Track cancellations and returns by category, fulfilment, and shipping level.
• Compare category order volume with sales contribution and average amount.
• Investigate missing Amount values before calculating revenue KPIs.
• Compare delivery outcomes for Amazon and Merchant fulfilment.
• Use geographic patterns to support inventory and delivery planning.

Conclusion

The dataset supports analysis of order activity, product demand, fulfilment, shipping preferences, and geography. A reproducible Python EDA workflow and a Power BI dashboard can make these patterns easier to explore. Clear documentation of data-cleaning decisions and order-status rules is essential for trustworthy business conclusions.

Repository Suggestion

README.md | notebooks/amazon_sales_analysis.ipynb | images/dashboard.png | reports/Amazon_Sales_Analysis_Report.pdf | data/README.md

Author: Saif Ali Khan | Project: Amazon Sales Analysis